Mechanical Arm Paper

Working Draft — Closest-Work Update v1.1 (based on Frozen v1)

初稿｜2026-09-28｜Sections 2–3 updated after targeted closest-work audit

Document status
本文件是在 Frozen Research Positioning & Manuscript Integration v1 基础上形成的 closest-work update 初稿。Sections 2–3 为 Working Draft v1.1，尚未重新冻结；其余未修改部分继承 v1 的冻结状态。本轮只更新 novelty / closest-work / contribution positioning，不新增实验结论。

最高原则：从现在起，论文不再围绕“哪个方法更准”或“高度超过多少必须 3D”组织，而围绕 model adequacy 展开：一个几何上并不完全正确的 planar localisation model，是否仍然足以满足下游任务容差？

1. Frozen Paper Identity

1.1 Primary title

Recommended title
When Is 3D Worth It? Task-Conditioned Adequacy of Planar Localisation for Vision-Guided Manipulation

正式备选标题：Task-Conditioned Adequacy of Planar Localisation under Off-Plane Model Mismatch。前者更适合作为应用型机器人视觉论文标题；后者更偏方法学。

1.2 Intellectual center

Core idea
Model correctness is not the same as model adequacy. A planar model can be geometrically imperfect outside its calibration plane yet remain adequate for a task if the induced localisation error stays within the task tolerance.

因此，论文不再把“场景是否严格平面”作为最终判据，也不把“3D 是否更准确”直接等价为“3D 是否值得使用”。真正的工程判据是：模型失配所造成的定位误差是否已经大到会影响当前任务。

1.3 Formal research questions

Primary RQ (frozen):

Primary RQ
Under what geometric and bounded-error conditions does a planar image-to-workspace localisation model remain adequate for a specified manipulation tolerance, despite off-plane model mismatch?

Secondary RQ:

Secondary RQ
When does a 3D-aware localisation method become necessary to obtain that guarantee?

通俗 framing 可以继续使用 “When is planar 2D mapping sufficient, and when is 3D worth it?”；但正式论证必须始终回到 task-conditioned adequacy。

2. Novelty Repositioning — Working Draft v1.1

2.1 What is NOT claimed as novelty

- Homography / planar mapping 在离开参考平面后产生 projective bias 或 parallax。
- Off-plane error 的 h/(H-h) 类 projective structure，以及对该 geometry-induced bias 的物理 bound。
- Localization error 应相对于 downstream task requirement / tolerance 判断这一上位思想。
- Homography uncertainty / covariance propagation；unknown-but-bounded / set-membership localisation。
- Task-oriented calibration，或“哪些 error components 对 task 有意义”的一般思想。
- Homography vs PnP comparison；planar/non-planar model selection；2D/3D representation switching。

- 根据 desired localisation accuracy 进行 sensor placement / minimum sensing，或 task-driven sensor / algorithm co-design。
- Operating-envelope / worst-case bound 这个抽象概念。

2.2 Revised minimum indivisible novelty unit

Working novelty nucleus
Task-conditioned adequacy of calibration-plane image-to-workspace localisation under analytically predictable off-plane model mismatch and separately bounded real-system residual error, with an explicit distinction between 3D necessity for a task guarantee and 3D accuracy dominance.

$E_{geom}(h, \rho, C_z) + S_P \le \tau$

这里不再把任何 constituent idea 单独包装成 novelty。当前可守的贡献在于：在 calibration-plane image-to-workspace localisation 这个具体问题里，把 analytically predictable off-plane mismatch 与 geometry 之外的 bounded real-system residual 分开，再把二者的组合相对于 manipulation tolerance 解释为 model adequacy；同时把"3D 为获得 task guarantee 所必需"与"3D 只是 robustly more accurate"分成两个不同命题。

Working novelty sentence
Prior work separately covers bounded off-plane projection error, task-specific localisation requirements, planar/non-planar model or representation selection, and bounded-error localisation. This work focuses on their combination in calibration-plane image-to-workspace localisation: analytically predicted off-plane model mismatch is kept distinct from bounded real-system residual error, and the combined bound is evaluated against a manipulation tolerance to separate planar sufficiency, 3D-required-for-guarantee, and 3D accuracy-dominance regimes.

2.3 Closest-work audit criteria

A — 解析刻画 planar assumption 在 off-plane / non-planar 条件下产生的几何 mismatch。

B — 除几何 mismatch 外，显式处理 bounded real-system uncertainty / residual。

C — 存在明确的 downstream task tolerance / requirement。

D — 涉及 planar/simple 与 richer/3D model 的选择、切换或必要性判断。

E — 给出 bound、operating envelope 或 guarantee / feasibility condition。

F — 有真实物理系统验证；"external"表示另有独立系统/数据的结构性支持。

2.4 Closest-work matrix — targeted audit

Work	A	B	C	D	E	F
Flatland Revisited (Carlson, 2025)	✓	✗	✓	◐	✓	✓
SLP homography-bias work (Liu et al., 2023)	✓	✓*	◐	✗	✓	✓
Selective Fiducials (Carlson & Kolling, 2023)	✗	◐	✓	✗	✓	◐
ORB-SLAM (Mur-Artal et al., 2015)	◐	◐	✗	✓	✗	✓
ADAMS (Imad & Shin, 2026)	✗	✗	◐	✓	✗	✓
Task-oriented calibration (Song et al., 2023)	✗	◐	✓	✗	✗	✓
Guaranteed interval localisation (Guyonneau et al., 2014)	✗	✓	◐	✗	✓	✓
UWB sensor placement (Zhao et al., 2022)	✗	◐	✓	◐	✓	✓
CODEI (Milojevic et al., 2025)	✗	◐	✓	◐/✓	◐/✓	◐
Current work	✓	✓ scenario	✓	✓	✓	✓ + external

Legend：✓= clearly covered；◐= partial / indirect；✗= not a central component。 SLP 中 bounded 的主要是 geometry-induced cross-modal mapping bias，不等于本论文中 geometry 之外的完整 real-system residual S_P。*

2.5 Highest-threat comparisons and remaining distinction

Closest-work group	Already covers	Remaining distinction for this paper
Flatland Revisited	Flat/non-flat assumption; analytic worst-case	Does not separate geometry-induced mismatch

Closest-work group	Already covers	Remaining distinction for this paper
	image-feature pose error; explicit 5 cm task requirement; richer non-flat estimator; robot field validation.	from an independently bounded real-system residual; does not explicitly distinguish "3D required for a guarantee" from "3D more accurate".
SLP homography-bias work	Closed-form elevation-induced homography mapping bias and a physical bound on that mapping bias.	Its bounded term is geometry-induced cross-modal mapping bias, not the complete real-system residual represented by S_P in our framework.
Selective Fiducials	Localization error is judged against task-dependent requirements; localization tasks/fiducial placement are designed to reduce task failure.	Selection variable is localization support / fiducial placement, not the adequacy of a planar image-to-workspace model under off-plane mismatch.
ORB-SLAM / ADAMS	Planar-vs-richer geometric model or 2D/3D representation selection/switching already exists.	Their switching/selection is driven by scene geometry/model fit or structural/resource considerations, not by an explicit manipulation-tolerance guarantee for a known planar-model mismatch.
Bounded-error / sensor-selection / CODEI families	Bounded localization guarantees, desired-accuracy-driven sensing, and task-driven perception complexity are established ideas.	They do not target the specific decomposition: analytic off-plane planar-model mismatch + separate real-system residual + manipulation tolerance + necessity-vs-dominance semantics.

No.1 problem-level neighbor — Flatland Revisited

Carlson (ICRA 2025 Robots in the Wild Workshop) 已经把 non-flat floor assumption、解析 worst-case image-feature pose error、显式 5 cm task accuracy requirement、richer non-flat estimator 和真实机器人部署连在一起。因此，"把 planar-assumption error 与 task tolerance 联系起来"本身不能再作为本论文 novelty。它目前是最需要在 Related Work 正面比较的工作。

仍可守的区别：本论文显式把 E_geom 与独立的 real-system residual scenario S_P 分开，并进一步建立 planar guarantee / 3D guarantee / robust error dominance 的不同语义；Flatland 主要解决一个具体 flat-floor failure mode，并未形成这一套 residual separation 与 necessity-vs-dominance framework。

Geometry-level neighbor — SLP

Liu et al. 的 SLP work 已经给出 elevation-induced homography mapping bias 的闭式几何关系，并基于物理范围给出 bounded mapping bias。因此本论文不能 claim off-plane projective-bias formula / bounding 本身。我们的剩余区分是：把 geometry-induced mismatch 与 geometry 之外的 real-system residual 分开，并把组合误差用于 manipulation-task adequacy。

Task-requirement neighbor — Selective Placement of Visual Fiducials

Carlson & Kolling (ICRA 2023 Workshop) 明确提出 localisation performance 应根据不同 task 的 localisation requirement 判断，并通过 fiducial placement 降低 task failure。因此 "task-dependent localisation requirement" 也不是新思想。我们的 selection variable 不是 fiducial / sensing support，而是一个已知存在 off-plane mismatch 的 planar image-to-workspace model 是否仍然足够。

Model-selection / co-design neighbors

ORB-SLAM 已经在初始化阶段并行考虑 homography 与 fundamental matrix，并按 geometric model support 选择恢复方式；ADAMS 已经根据 scene structural complexity 在 2D / 3D mapping 间切换；Zhao et al. 和 CODEI 进一步说明 desired accuracy / downstream requirements 可以驱动 sensing 或 perception-stack 选择。因此本论文不能 claim "planar-vs-3D selection" 或 "task-driven perception complexity" 本身。可守的区别仍是具体的 error decomposition + manipulation-tolerance guarantee semantics。

2.6 Novelty audit status — working conclusion

Targeted-search verdict
经过当前 targeted closest-work audit，尚未找到一项工作在 vision-guided manipulation localisation 的具体问题中同时覆盖完整链条：
(1) 解析预测 calibration-plane mapping 的 off-plane error；
(2) 将该 model mismatch 与独立 bounded real-system residual 分离；
(3) 相对于显式 manipulation tolerance 建立 model-adequacy guarantee；
(4) 区分 3D 的 task necessity 与 accuracy dominance；
(5) 通过受控物理实验并辅以 independent external structural validation 支撑机制。

这不是 exhaustive proof of novelty，也不能写成 "first ever"。当前最稳妥的定位仍是 applied methodological / problem-formulation contribution。

Manuscript guardrail：不要再写 "No prior work has studied task-conditioned planar adequacy"。Related Work 应主动把 Flatland Revisited 放在核心邻居位置，然后明确说明我们从它已经覆盖的 task-threshold logic 往前推进到 residual separation、guarantee semantics 与 manipulation-specific model adequacy。

2.7 Closest-work references used in this draft

[CW1] Carlson, J. (2025). Flatland Revisited: The Effects of Non-Planar Floors on Large-Scale Warehouse Deployment of Autonomous Mobile Robots. ICRA 2025 Robots in the Wild Workshop.

[CW2] Liu, S., Huang, X., Fu, N., Li, C., Su, Z., & Ostadabbas, S. (2023). Simultaneously-Collected Multimodal Lying Pose Dataset: Enabling In-Bed Human Pose Monitoring. IEEE TPAMI, 45(1), 1106–1118. DOI: 10.1109/TPAMI.2022.3155712.

[CW3] Carlson, J., & Kolling, A. (2023). Selective Placement of Visual Fiducials for Localization Tasks. ICRA Workshop on Robot Execution Failures and Failure Management Strategies.

[CW4] Mur-Artal, R., Montiel, J. M. M., & Tardós, J. D. (2015). ORB-SLAM: A Versatile and Accurate Monocular SLAM System. IEEE Transactions on Robotics, 31(5), 1147–1163. DOI: 10.1109/TRO.2015.2463671.

[CW5] Imad, M., & Shin, S. Y. (2026). Scene recognition-based adaptive map switching for resource-constrained robotic navigation. Engineering Applications of Artificial Intelligence, 177(P2), 115005. DOI: 10.1016/j.engappai.2026.115005.

[CW6] Song, Y., Tian, W., Tian, Y., & Liu, X. (2023). A task-oriented calibration method for a 5-DOF hybrid machining robot based on the equivalent system. Measurement, 112909. DOI: 10.1016/j.measurement.2023.112909.

[CW7] Guyonneau, R., Lagrange, S., Hardouin, L., & Lucidarme, P. (2014). Guaranteed Interval Analysis Localization for Mobile Robots. Advanced Robotics, 28(16). DOI: 10.1080/01691864.2014.908742.

[CW8] Zhao, W., Goudar, A., & Schoellig, A. P. (2022). Finding the Right Place: Sensor Placement for UWB Time Difference of Arrival Localization in Cluttered Indoor Environments. IEEE Robotics and Automation Letters, 7(3), 6075–6082. DOI: 10.1109/LRA.2022.3165181.

[CW9] Milojevic, D., Zardini, G., Elser, M., Censi, A., & Frazzoli, E. (2025). CODEI: Resource-Efficient Task-Driven Co-Design of Perception and Decision Making for Mobile Robots Applied to Autonomous Vehicles. IEEE Transactions on Robotics, 41, 2727–2748. DOI: 10.1109/TRO.2025.3552347.

3. Contributions — Working Draft v1.1

以下 wording 已根据 closest-work audit 收窄。Contribution 2 仍是主 applied methodological contribution；Contribution 1 和 3 提供 context-specific analytical / empirical support。三条均不得解释为 constituent ideas 的 first-ever claim。

Contribution 1 — Context-specific geometry-to-error characterization

We derive the calibration-plane image-to-workspace form of the off-plane localisation bias in terms of relative target height and lateral camera–target geometry, and validate its dominant magnitude, direction, and spatial structure using controlled physical experiments without fitting the nonzero-height observations.

Contribution 2 — Main applied methodological contribution

We formulate calibration-plane image-to-workspace localisation as a task-conditioned adequacy problem by separating analytically predictable off-plane projective mismatch from a bounded real-system residual term and comparing their combined bound with an explicit manipulation tolerance. The resulting framework distinguishes

planar sufficiency, 3D necessity for a task guarantee, and 3D robust error dominance as separate conditions rather than treating them as equivalent.

Contribution 3 — Multi-layer empirical validation

We provide a multi-layer empirical validation of this formulation. Controlled experiments across repeated physical rebuilds and two cameras validate the principal geometric mechanism; an uncertainty audit shows that independently recoverable measurement variability does not explain the full residual; and an independent STAR-loc analysis provides supporting cross-system evidence that the structural projective mechanism is not specific to the original low-cost camera setup.

Contribution hierarchy
C2 = main applied methodological / problem-formulation contribution.
C1 = context-specific analytical + physical contribution; do not claim the projective form itself is fundamentally new.
C3 = empirical / validation contribution.

Publication-facing rule: novelty is the integrated formulation and semantic separation, not "six columns all checked".

4. Evidence Architecture — Frozen

Evidence layer	Question answered	Current status	Paper role
L1 Analytical mechanism	为什么 planar 离面会产生系统性偏差？	Complete / very strong	Main evidence
L2 In-plane baseline (E1)	planar 方法在自己的适用平面是否成立？	Complete / strong	Supporting baseline
L3 Controlled physical validation (E2)	理论能否预测真实离面退化？	Complete / very strong	Main evidence
L4 Residual & uncertainty audit	剩余误差能否被已测 measurement variability 解释？	Complete / strong	Supporting evidence
L5 Task-conditioned adequacy	何时 planar 保证够用，何时只有 3D 有 guarantee？	Complete / very strong conditional result	Main methodological result
L6 External structural validation (STAR-loc)	mechanism 是否只是 Berxel/E2 特例？	Complete / strong	Supporting external evidence
L7 Literature scenario plausibility	S_P / S_3D 场景范围是否现实得离谱？	Complete / moderate	Context / bounds

Evidence sufficiency verdict：EVIDENCE SUFFICIENT FOR CURRENT CLAIM SCOPE。当前不建议主动增加第三个 dataset、新 RGB-D、新 robot end-to-end task、新 depth model 或新的 calibration algorithm。

5. Final Claim–Evidence Matrix

ID	Frozen claim	Primary evidence	Strength	Forbidden stronger wording
C1	Planar off-plane bias follows an explicit projective geometry law.	Analytical derivation	Very strong	"所有真实误差都由 geometry 决定"
C2	Z=0 geometry predicts the dominant magnitude/direction/spatial structure of E2 off-plane degradation without fitting h>0.	Stage 2 / E2	Very strong	"完整系统 accuracy 被精确预测"
C3	Independently recoverable pixel/pose/map variability is insufficient to explain the remaining real residual.	Stage 3	Strong	"所有 measurement/setup error 都不重要"
C4	Empirical PnP is less height-sensitive than frozen planar mapping, but its absolute accuracy is not explained by the simple measurement model.	Stage 3 PnP	Strong, bounded	"PnP 最优 / PnP uncertainty fully validated"
C5	Planar localisation is guaranteed sufficient under declared bounds when E_geom + S_P ≤ τ.	Stage 4B	Very strong conditional	"否则 planar 一定失败"

ID	Frozen claim	Primary evidence	Strength	Forbidden stronger wording
C6	3D-only guarantee and 3D error dominance are distinct concepts.	Stage 4B	Very strong conditional	“3D 更准 = 任务必须 3D”
C7	External evidence supports the declared error ranges as broad sensitivity scenarios, not universal empirical bounds.	Stage 5	Moderate supporting	“literature validated S_P=10 / S_3D=15 as universal caps”
C8	STAR-loc supports cross-system generalisation of the structural projective mechanism.	Stage 2.1 STAR-loc	Strong supporting	“STAR-loc independently validates our absolute accuracy / task boundary”

6. Frozen Mathematical Spine and Decision Semantics

6.1 Off-plane projective geometry

$$\Delta X = h(X - C_x)/(C_z - h), \quad \Delta Y = h(Y - C_y)/(C_z - h)$$

$$E_{geom} = |h|\,\rho\,/\,|C_z - h|, \quad \rho = \sqrt{[(X - C_x)^2 + (Y - C_y)^2]}$$

$$\alpha = h/C_z, \quad \beta = \rho/C_z, \quad E_{geom}/C_z = |\alpha|\beta/|1 - \alpha|$$

Interpretation：离面误差不是由“高度多少毫米”单独决定，而由 relative height 与 lateral camera-target geometry 共同决定。固定 20 mm / 50 mm / 5 cm crossover 均不得写成 universal result。

6.2 Bounded planar error

$$E_P^{lower} = \max(0, E_{geom} - S_P), \quad E_P^{upper} = E_{geom} + S_P$$

S_P 的正式含义：planar systematic-error magnitude bound / scenario。它不是 precision、不是 Gaussian σ，也不是从 Stage 3 residual 反推出来的真实参数。

6.3 Task-conditioned adequacy

$$G_P = [E_{geom} + S_P \le \tau]$$

$$G_{3D} = [S_{3D} \le \tau]$$

Planar	3D-aware	Frozen interpretation
Guaranteed	Guaranteed	Both guaranteed; 3D may only provide optional accuracy gain
Not guaranteed	Guaranteed	3D required for a guarantee
Guaranteed	Not guaranteed	Planar only guaranteed under declared scenarios
Not guaranteed	Not guaranteed	Neither method currently guaranteed; this does not mean both fail

6.4 Error dominance is independent

$$THREE_D_ROBUST_ERROR_DOMINANCE \Leftrightarrow E_{geom} > S_P + S_{3D}$$

Semantic guardrail
3D robustly more accurate ≠ 3D required by the task. Error dominance and task necessity must remain separate in every figure, table and paragraph.

6.5 Operating-envelope form

$$E_{geom}^{max} = h_{max}\,\rho_{max} / (C_z - h_{max})$$

E_geom^max + S_P ≤ τ ⇒ planar adequate over the declared envelope

此 operating-envelope form 是 practical presentation，不作为独立 novel algorithm / "Planar Sufficiency Certificate" 新算法进行宣传。

7. Key Quantitative Evidence to Preserve

7.1 E1 planar baseline

- Formal held-out Homography mean / RMSE / max ≈ 0.585 / 0.660 / 1.197 mm。
- PnP held-out ≈ 0.587 / 0.661 / 1.293 mm；Affine 显著更差。

作用：只证明 planar mapping 在 calibration plane 上可以工作；不把 E1 当 off-plane novelty。

7.2 E2 geometry validation

- 5 physical rebuilds × 2 cameras；height 0–50 mm。
- Height-curve correlation ≈ 0.9942–0.9951。
- Point-level magnitude correlation ≈ 0.9869–0.9942。
- Median direction cosine ≈ 0.9983–0.9992。
- Geometry-unexplained vector RMSE ≈ 3.61–4.36 mm。

冻结 wording：Under the tested fixed-board E2 protocol, off-plane planar-mapping degradation was largely predictable from Z=0 projective geometry。

7.3 Stage 3 residual / uncertainty audit

- Planar measurement-only radial Q95 roughly 0.024–0.266 mm at σ_h=0 scenario。
- Real geometry-unexplained residual RMSE ≈ 3.61–4.36 mm，存在 directional / rebuild / height structure。
- PnP empirical XY RMSE ≈ 9.99 mm (ihawk1) / 5.10 mm (ihawk2)，measurement-only model under-covers reality。

冻结结论：Known independently recoverable pixel/pose/map variability is insufficient to explain the full real residual。不要把 residual 简化成 sensor noise，也不要反推真实 σ_h。

8. STAR-loc External Structural Validation — Frozen Role

Formal identity
Supporting external validation of the structural projective mechanism.

STAR-loc 不是第二个主实验、不是 E2 replication、不是 absolute accuracy 独立验证、不是 S_P estimation dataset，也不是第二套 planar-vs-3D task-boundary 实验。

8.1 Methods elements retained in main paper

- 8 held-out setup-s3 sequences；apriltag_s3 calibration sequence excluded from evaluation。
- Globally calibrated camera pose + real measured AprilTag pixels。
- Externally reconstructed reference plane from known 3-D low-height landmark subset；image residuals 不用于拟合 reference plane。
- Define e_geom, e_obs and residual r = e_obs − e_geom。
- No off-plane observations are used to fit or rescale the projective prediction。

8.2 Main external results retained

- Retained observations: 12,473 repeated observations across 55 tags and 8 sequences（不能写成 12,473 independent physical samples）。
- Spearman magnitude ordering $\rho_s \approx 0.981$。
- Median direction cosine ≈ 0.998；direction agreement is signal-dependent。
- $P(R<D) \approx 88.65\%$；$P(R<0.5D) \approx 75.54\%$；median $R/D \approx 0.183$。
- Rare catastrophic tail：top 13 rows (~0.1%) contribute ~99.79% of residual SSE；largest failures correspond to near-grazing measured rays / ill-conditioned ray-plane intersections。

8.3 Absolute-scale limitation

STAR-loc 的 low-landmark reference plane 与 Vicon-horizontal definition 存在不可忽略差别，因此 absolute-scale generalisation 不能被干净识别。正文最多作为 limitation；不把 reference-plane difference 本身包装成新 phenomenon。

8.4 Paper placement

正文只保留：1 个 supporting subsection + 1 个 3-panel figure + 1 个 compact table；完整 heavy-tail、conditioning、plane sensitivity、matched-domain、macro analyses 全部进入 Supplementary / Appendix。

9. External Literature & Scenario Plausibility — Frozen Interpretation

Stage 5 的正确作用不是“external evidence calibrated our bounds”，而是判断 declared scenario ranges 是否适合作为 broad sensitivity analysis。

Quantity	Frozen scenario set	Allowed interpretation	Forbidden interpretation
S_P	0, 1, 2, 3, 4, 5, 7.5, 10 mm	Broad planar systematic-error scenarios; low-single-mm controlled outcomes are externally plausible	Universal planar error upper bound / population Q95
S_3D	1, 2, 3, 5, 7.5, 10, 15 mm	Broad controlled-domain 3D-aware XY error scenarios	Universal PnP cap; adverse-view literature can greatly exceed 15 mm

Frozen verdict：Broad scenario range externally plausible, not calibrated；no justified scenario-set update；no new acquisition required for the current conditional claims。

Accuracy / precision boundary：所有 Stage 4B decision quantities 必须使用 board/work-plane XY Euclidean localisation error。Precision / repeatability 不能代替 ground-truth accuracy；3-D translation / range / rotation / reprojection metrics 不得硬转成 XY bound。

10. Terminology Freeze — Mandatory Replacements

Avoid / legacy wording	Use instead
2D/3D selection	task-conditioned model adequacy
3D advantageous	3D robust error dominance
3D necessary	3D required for a guarantee
planar fails	planar is not guaranteed under the declared bound
crossover	task-conditioned boundary / operating regime
uncertainty model (when not probabilistic)	bounded real-system error scenario

Avoid / legacy wording	Use instead
3D error floor	3D-aware XY error upper-bound scenario
success probability	scenario fraction / guarantee class
external validation of accuracy (STAR-loc)	external structural validation
calibration selection	localisation-model adequacy
universal threshold	declared operating boundary

Non-negotiable wording rule
"Not guaranteed" does not mean "fails"; "3D more accurate" does not mean "3D required"; scenario fractions are not success probabilities.

11. Manuscript Structure — Frozen v1

Section	Purpose / question answered	Primary evidence
1. Introduction	Why does model adequacy matter in nominally planar manipulation? What is the gap?	Closest-work positioning + C1–C3
2. Related Work	Separate geometric model validity, task-oriented accuracy, bounded-error localisation; end at model-adequacy gap	Targeted closest work
3. Problem Formulation	Define h, ρ, C_z, τ, S_P, S_3D and common XY metric	Definitions
4. Geometry Model	Derive off-plane bias and normalized variables	C1
5. Experimental Evidence	E1 baseline + E2 controlled held-out-height validation	C2
6. Residual & Measurement-Uncertainty Audit	Show geometry explains dominant effect but not full residual	C3/C4
7. Task-Conditioned Model Adequacy	Define G_P, G_3D, four guarantee classes, separate error dominance	C5/C6
8. Results / Operating Regimes	Show task-conditioned maps + E2 projection	Stage 4B
9. External Structural Validation	Compact STAR-loc supporting module	C8
10. Discussion	Correctness vs adequacy; dominance vs necessity; no universal crossover; external scope	All
11. Limitations	n=5 rebuilds, scenario bounds, PnP reference, XY scope, conditional guarantees	Claim boundaries
12. Conclusion	Need for 3D depends on task-relevant induced error, not departure from planarity alone	Core takeaway

12. Main Figure / Table Freeze

12.1 Main figures

1. Figure 1 — Problem-definition / adequacy concept: planar calibration → off-plane mismatch → compare E_geom + S_P with τ → PLANAR ADEQUATE / NOT CERTIFIED.
2. Figure 2 — E2 blind held-out-height geometry validation: predicted vs observed magnitude / height curves across rebuilds and cameras.
3. Figure 3 — What geometry does not explain: geometry residual versus measurement-only uncertainty / residual structure.

4. Figure 4 — Task-conditioned adequacy regimes: BOTH GUARANTEED / 3D ONLY / PLANAR ONLY / NEITHER; do not reuse Stage 4A mixed semantics.
5. Figure 5 — STAR-loc external structural validation: magnitude structure; direction vs signal; residual vs measured-ray conditioning.

12.2 Main tables

6. Table 1 — Dataset / evidence overview: E1, E2, Stage 3, STAR-loc, role and evidence grade.
7. Table 2 — Stage 2 main quantitative validation metrics.
8. Table 3 — Task-aware decision conditions and interpretations.

Stage 5 external evidence 建议作为 compact Discussion / Supplementary table，不占用主 Results figure quota。

13. Scope Boundaries and Explicit Non-Claims

- 不声称存在 universal 20 mm / 50 mm / 5 cm crossover。
- 不声称 3D always better / always necessary。
- 不声称 PnP 是最优 3D 方法。
- 不声称 S_P 或 S_3D 是经过 population calibration 的 universal bounds。
- 不声称 Stage 3 residual 的真实 causal source 已被识别。
- 不声称 manufacturer depth specification 等价于 statistical σ_z。
- 不声称当前结果是完整 robot grasp / manipulation success guarantee。
- 不声称 STAR-loc 对 absolute accuracy、S_P 或 Stage 4B boundary 做了独立 replication。
- 不把 scenario fraction 解释为 probability / likelihood。
- 不把 operating-envelope form 包装成一个新算法或 novel theorem。

14. Limitations — Frozen

Limitation	Correct interpretation / mitigation
E2 independent rebuilds n=5	足以支持受控 mechanism validation；不作为 population-level distribution estimate
S_P / S_3D are scenarios	所有 guarantee 都是 under declared error scenarios，而非 universal probabilistic guarantee
PnP is one 3D-aware reference	用于说明 height robustness / comparative role，不代表所有 3D methods
Primary metric is board/work-plane XY	不扩展为 full SE(3) pose、grasp success 或 end-to-end robot accuracy
STAR-loc reference plane is externally reconstructed	只支持 structural mechanism generalisation；absolute-scale interpretation 有边界
Closest-work audit is targeted, not systematic	避免 first-ever wording；使用 “we formulate / we establish / we show”

15. Current Project Maturity

Scientific research completeness

Working estimate: ~93–95% for the current frozen claim scope. The core mechanism, controlled validation, uncertainty audit, task-conditioned adequacy framework, external structural validation and scenario-plausibility audit are all present.

Manuscript completeness
Working estimate: ~70–75%. Remaining work is mainly paper integration, final figure generation, source/citation assembly, paragraph-level writing, supplementary packaging and reviewer-style audit.

当前最大的风险已经不是"缺一个实验"，而是 manuscript 写作时发生 claim inflation、旧术语回流或把不同证据层重新混在一起。

16. Next Step — Final Paper Blueprint v1

下一步不再新增研究 Stage。直接建立 paragraph-level manuscript blueprint：Section → subsection → paragraph purpose → frozen claim → evidence → figure/table → wording boundary → source/citation need。

Priority	Action	Deliverable
1	Final Paper Blueprint v1	逐段写作蓝图，覆盖 Introduction → Conclusion
2	Citation / closest-work assembly	为 Related Work 和 novelty gap 绑定最终高质量来源
3	Final Figure Pack	冻结 5 张主图 + supplementary figures
4	Claim traceability audit	每条正文 claim 指向 equation / dataset / figure / limitation
5	Full manuscript drafting	按蓝图逐节写，避免再次扩题
6	Reviewer-style adversarial audit	检查 novelty inflation、circular validation、probability/guarantee、accuracy/precision、metric mismatch

Stop rule
除非 reviewer-level audit 发现一个会改变核心结论的缺失证据，否则不再主动增加新实验、新 dataset、新 depth branch 或新算法。

Appendix A. Repository Stage → Paper Role Crosswalk

Repository stage	Frozen paper-facing identity
Stage 1	Data provenance and evidence audit
Stage 2	Off-plane geometric model + controlled physical validation
Stage 3	Residual and measurement-uncertainty audit + partial PnP reference
Stage 4A	Robust error-bound analysis (supporting derivation; legacy mixed semantics not reused)
Stage 4B	Task-conditioned model adequacy (primary decision framework)
Stage 5	External scenario plausibility / bounds context
Stage 2.1 STAR-loc	Supporting external structural validation

Appendix B. Frozen One-Sentence Takeaways

- Geometry takeaway: planar off-plane error depends on relative height and lateral camera-target geometry, not height alone.
- Empirical takeaway: Z=0 geometry predicts the dominant E2 off-plane degradation without fitting nonzero heights.

- Residual takeaway: the remaining real residual is larger and more structured than independently recoverable frame-level measurement variability.
- Decision takeaway: a geometrically imperfect planar model can still be task-adequate.
- 3D takeaway: 3D being more accurate is not the same as 3D being required.
- External takeaway: STAR-loc supports mechanism generalisation, not absolute-accuracy replication.
- Scenario takeaway: S_P and S_3D remain broad declared sensitivity scenarios, not universal empirical constants.
- Paper takeaway: the need for 3D localisation is determined by task-relevant induced error, not departure from planarity alone.